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Keywords

artificial intelligence, computing and processing, deep learning, machine learning, precision medicine, survey

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A Comprehensive Survey of AI Models in Precision Medicine

Zhixing Wang Hao Tang*

***Abstract*— In this paper, we present a comprehensive overview of precision medicine through the lens of artificial intelligence (AI), emphasizing how computational technologies are reshaping modern healthcare. To provide a thorough and up-to-date perspective, we synthesize insights from over 275 relevant publications spanning key areas of AI application. Precision medicine seeks to optimize diagnosis and treatment by integrating individual biological and clinical data to deliver more targeted and effective medical interventions. This intersection has become increasingly significant due to the rising complexity of clinical data and the growing demand for personalized, data-driven medical solutions. Specifically, we review the integration of AI—particularly machine learning and deep learning—into core domains such as multi-omics analysis, personalized drug discovery, physiological signal interpretation, treatment response prediction, and biological pathway modeling. Furthermore, we examine AI applications in optimizing treatment regimens, accelerating molecular synthesis, and enhancing medical image analysis through automated feature extraction and segmentation techniques. Across these areas, AI contributes to improving diagnostic accuracy, identifying therapeutic targets, predicting clinical outcomes, and generating personalized treatment strategies based on patient-specific characteristics. These rapid developments highlight AI's increasing potential to transition precision medicine from research environments to real-world clinical practice, improving efficiency and scalability. By systematically outlining current applications and their transformative impact, this review provides a structured reference for researchers, clinicians, and policymakers involved in AI-assisted precision healthcare. Looking ahead, continued progress in algorithm interpretability, data integration, and ethical frameworks will be essential for realizing the full clinical potential of AI-enabled precision medicine.**

***Index Terms*— Precision Medicine, Artificial Intelligence, Machine Learning , Deep Learning , Survey**

I. Introduction

Precision medicine is commonly defined as an approach that improves upon traditional treatment selection by tailoring interventions to individual characteristics and needs which presents a schematic overview of precision medicine. It involves the use of individual features, computational power, and algorithms to predict and optimize individual disease risk and treatment response for healthcare and clinical decision-making, thus making treatments safer, more effective, and more cost-effective [1]–[4].

In 2015, U.S. President Barack Obama formally launched the Precision Medicine Initiative (PMI), a national effort aimed at transforming healthcare by tailoring treatment to individual variability. The initiative brought precision medicine into the public spotlight, with a call to "cure cancer and diabetes" significantly accelerating both scientific research in this field [5]. Since then, realizing the vision of precision medicine has relied heavily on the advancement of computational technologies and large-scale biomedical data infrastructure. With the expansion of computing power, a foundation has been built for our analytical capabilities and the ability to link data sets, translating them into biomarkers [6], [7]. Technological advancements and a massive volume of clinical observational data have become available to inform the practice of precision medicine [8], [9].

To provide a structured overview of current progress in medical artificial intelligence, this review summarizes recent advances across several key domains.

- Modeling. The development of graph neural networks and generative models has significantly advanced molecular representation and compound design in drug discovery. These models effectively capture molecular structures and chemical properties, contributing to improvements in target identification, virtual screening, and molecular property prediction, with increasing alignment to biological relevance.
- Signal Processing. Deep learning approaches have demonstrated robust capabilities in analyzing biomedical signals such as ECG and EEG. Notable progress includes effective noise reduction, real-time arrhythmia detection, and multimodal signal fusion, which have collectively improved the accuracy and efficiency of physiological monitoring and early disease detection.
- Multimodal Diagnosis. Transformer-based architectures and benchmark datasets have facilitated significant breakthroughs in clinical diagnosis by integrating structured records, medical imaging, and unstructured clinical text. These multimodal systems have achieved high performance in clinical reasoning tasks, supporting more comprehensive and consistent diagnostic processes.
- Personalization. Artificial intelligence has been increasingly applied to support individualized treatment strategies through models that incorporate patient-specific profiles. Reinforcement learning and outcome-aware model-

Zhixing Wang is with the Faculty of Science, Universiti Malaya, Kuala Lumpur, Malaysia. E-mail: 22085457@siswa.um.edu.my

Hao Tang is with the State Key Laboratory of Multimedia Information Processing, School of Computer Science, Peking University, Beijing 100871, China. E-mail: haotang@pku.edu.cn

*Corresponding author: Hao Tang.

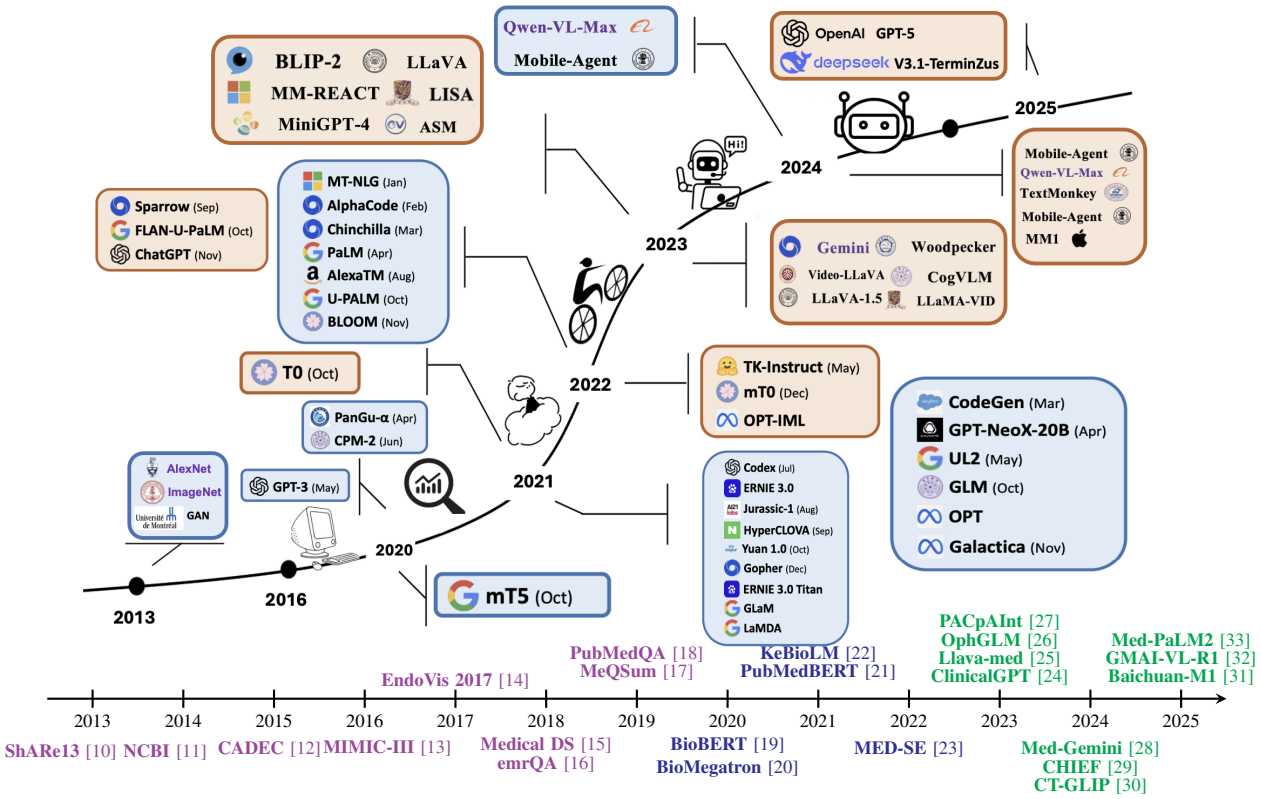


Fig. 1: The development of AI models in industry has progressively driven advancements in precision medicine. Starting from early image recognition models, through large language models, and then to molecular generation models, AI technologies have evolved step by step, each stage providing new tools and methodologies that have been adapted to biomedical applications. The data sets and benchmark models, Transformer-based pretrained models, and large language models (LLMs) and multimodal models are highlighted in purple, blue, and green, respectively [34]–[36].

ing have enabled adaptive treatment planning, enhancing the delivery of precision medicine and promoting more effective clinical decision-making.

By integrating these key areas, this review provides a clearer understanding of the evolving interface between artificial intelligence and medicine, and lays the groundwork for future integrative breakthroughs in both domains.

II. PRECISION MEDICINE

A. AI in Multi-Omics Analysis

Multi-omics encompasses valuable insights into human health, including disease-related information. As data volumes continue to increase, fields such as genomics, transcriptomics, proteomics, and metabolomics have emerged as the four main pillars of multi-omics [37]. This increasing data diversity presents both opportunities and challenges. To address the growing demand for data processing, AI technology has proven to be an efficient solution. By integrating data from various sources, AI enables researchers to gain a more comprehensive understanding of complex diseases and to develop novel therapeutic strategies [38]. Moreover, applying AI to multi-omics analysis accelerates our understanding of the biological and molecular mechanisms underlying these diseases, thereby driving advancements in precision medicine [39]. For example, transcriptomics can identify dysregulated gene pathways through gene expression data, while proteomics validates the proteins involved in these pathways. Metabolomics provides a dynamic snapshot of disease progression and treatment response, enabling real-time tracking [40]. Furthermore, comprehensive genomic profiling (CGP), via tissue-based and plasma-based assays, identifies key pathogenic and driver mutations, offering essential data for personalized treatment strategies [41]. To integrate such heterogeneous complementary data sources, AI-driven multi-omics models often learn joint representations across modalities. The fusion of multiple omics datasets can be represented by the following mathematical formulation. Next, this chapter will outline the current developments in multi-omics by focusing on proteomics and genomics, respectively.

1) Proteomics: Proteomics is the study of the complete set of proteins expressed by cells, tissues, or organisms at a given time, with the aim of elucidating their biological functions and processes [43], [44]. Although protein expression is initiated by transcription, this process is often accompanied by aberrations—such as alternative RNA splicing, various modifications, proteolytic processing, and post-translational modifications (PTMs) [45]—which can result in protein misfolding. Such aberrations may contribute to the onset of numerous diseases. For instance, in thalassemia, the absence of a specific protein precludes the production of functional hemoglobin, ultimately leading to disease development.

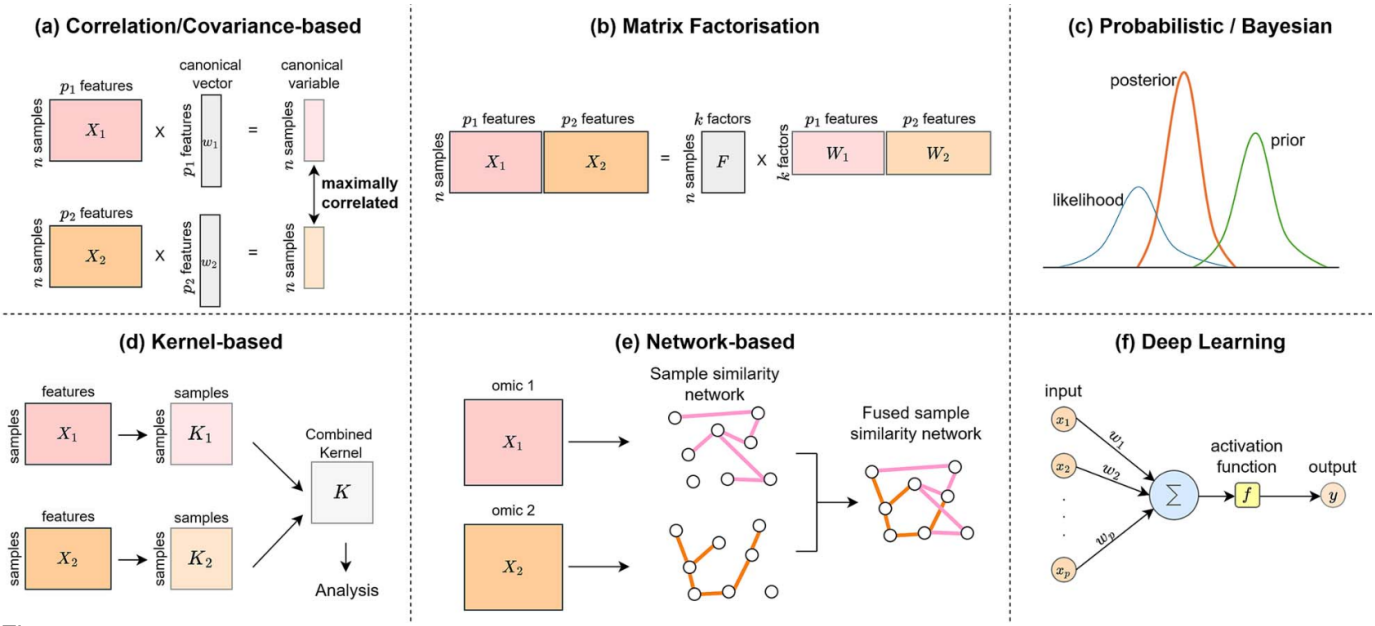


Fig. 2: Schematic diagram of multi-omics integration methods: (a) Canonical correlation analysis; (b) Matrix factorization that decomposes omics matrices into shared and omics-specific components; (c) Bayesian methods for probabilistic modeling; (d) Multiple kernel learning; (e) Similarity network fusion; (f) Artificial neuron — the basic unit of neural networks: For each sample vector, every input feature x is assigned a weight w; the weighted sum is then calculated, and the result is passed through an activation function to generate the neuron's output [42].

Traditionally, the subcellular localization of proteins has relied on wet lab techniques. Fluorescence microscopy, for example, employs fluorescent dyes or protein tags to label and visualize protein distributions within cells, while immunoelectron microscopy uses labeled antibodies to target proteins and is regarded as the gold standard for high-resolution imaging. Moreover, fluorescent biomarker tags, such as protein A-GFP [46], allow for the detection of protein signals across different cellular compartments. With technological advancements, the focus has gradually shifted from the localization of individual proteins to that of entire proteomes.

The primary proteomics detection techniques include mass spectrometry (MS)-based methods, which enable high-throughput protein analysis, along with techniques such as fluorescence-activated cell sorting (FACS) [47], cytometry by time of flight (CyTOF) [48], fluorescence imaging, and various immunoassays. These large-scale proteomic studies provide valuable insights into cellular functions and disease mechanisms. A key analytical step in such studies is the use of the log fold change ($\log_2$FC) to distinguish between disease and control conditions, as defined by the following formulation:

$$\log_2 \mathrm{FC}_i = \log_2 \left(\frac{E_i^{\text{condition B}}}{E_i^{\text{condition A}}} \right)$$

This quantitative metric highlights significant alterations in protein levels, enabling researchers to identify these proteins as primary biomarker candidates.

High-throughput proteomics, especially MS, facilitates the identification of protein biomarkers that can predict cardiovascular events more accurately than traditional risk factors, thereby aiding early disease detection and risk stratification. Proteomics further enhances our understanding of cellular processes, disease mechanisms, and treatment responses. Post-translational modifications are critical for protein functionality and help to reveal complex regulatory networks. By integrating proteomics with other omics layers, researchers can achieve a deeper understanding of how genetic variations and gene expression patterns influence protein activity, disease phenotypes, and overall cellular function. This holistic approach is essential for unraveling the intricacies of disease mechanisms and for identifying potential therapeutic targets [47], [49], [50].

One notable application is in oncology, where proteomics and related technologies quantify protein expression levels. Additionally, proteomics in liquid biopsy holds promise for developing individualized therapeutic strategies, particularly for ocular diseases [51]. In the realm of omics data, artificial intelligence has emerged as a fundamental tool for managing extensive and multidimensional datasets in biological analyses [52]. For instance, Go et al. developed the HepScope [53] gene set, a CNN-based framework that leverages protein data to identify relevant gene patterns. Enríquez-Rodríguez et al. [54] designed a method to forecast disease mortality probabilities based on proteomic features.Huang et al. [55] demonstrated the effectiveness of machine learning in identifying protein expression profiles and biomarkers associated with lung cancer. Nygaard et al. [56] developed an AI-based diagnostic model using proteomic data to identify biomarkers for multisystem inflammatory syndrome in children.

Overall, artificial intelligence is proving to be a powerful tool in the analysis of proteomics. By leveraging technologies such as CNNs and machine learning, researchers can now uncover critical biomarkers and predict disease outcomes with

greater accuracy. These advancements highlight AI's transformative potential in diagnostics and personalized medicine, deepening our understanding of complex diseases and facilitating the development of more effective therapeutic strategies.

2) Genomics: The development of genomics, driven by the rise of precision medicine, began with the discovery of the double-helix structure of DNA in 1953, advanced with the advent of DNA sequencing in 1977, and culminated in the Human Genome Project, which produced the first complete human genome in 2003 [57]. This progression underscores that complex diseases are influenced by genomic interactions; even highly similar genomes can lead to different diseases due to distinct mutations in various genes [58]. Currently, numerous computational models have been developed in the field of genomics. For instance, Wang et al. [59] proposed a personalized drug recommendation model that leverages genomic profiles to optimize therapeutic outcomes, thereby significantly aiding clinical treatment. Similarly, by integrating genomic data with drug response models, it is possible to identify genomic biomarkers that predict the effectiveness of cancer drugs [60] and to pinpoint cancer-specific DNA alterations to inform treatment decisions [61]. Moreover, computational models have advanced the extraction of genomic data; for example, Dubey et al. introduced the GenoAnalytica [62] model to analyze SARS genomic sequences, facilitating efficient data extraction from viral genomes.

Recent advances in artificial intelligence, particularly multimodal learning, have further integrated genomics with other omics data to provide precise information and support clinical decision-making. For example, Chang et al. [63] demonstrated that AI-driven analysis of the immune system combined with transcriptomics can effectively identify key genomic information, underscoring the significance of next-generation DNA and RNA sequencing tools in personalized therapeutic strategies. Similarly, Zeng et al. [64] proposed that integrating diverse omics data can drive technological innovation; notably, the fusion of radiomics and genomics has given rise to radiogenomics. This innovative approach merges artificial intelligence with biological insights to enable more comprehensive, precise, and personalized evaluations. Furthermore, the rapid advancement of both AI and omics technologies has led to significant applications in genomics, including tumor classification and subtyping, prognosis prediction, disease progression assessment, and the identification of abnormal genetic alterations.

In this section, AI has greatly advanced multi-omics by integrating complementary information [65] from different modalities, thereby enabling a deeper understanding of disease-related biological interactions and the heterophilic relationships of tissue-specific expression within biological networks [66], [67]. However, this field also faces significant challenges. The integration of multi-omics data using AI is complicated by issues such as the inherent high dimensionality and sparsity of single-cell data [68], which can lead to redundant and biased predictions with limited interpretability and predominantly low-level features [69].

Moreover, the absence of standardized integration frameworks further hampers the translation of these analyses into reliable clinical applications. Emerging methodologies that combine deep learning with advanced statistical techniques are showing promise in addressing these limitations, yet substantial research is still needed to refine these approaches and ensure consistent performance across diverse datasets. Addressing these challenges is crucial for further progress in multi-omics analysis, ultimately paving the way for more accurate diagnostics and personalized therapeutic strategies.

Method / Reference	SO	FE	MI
Structure and Feature Optimization			
FP-GNN [70]	✗	✓	✗
Multimodal-GNN [71]	✗	✓	✓
SLU-GNN [72]	✓	✗	✗
GKC [73]	✓	✓	✗
Hybrid-GNN [74]	✓	✗	✓
Structure Adaptation			
SSR-GNN [75]	✓	✓	✗
Spiers-GNN [76]	✗	✗	✗
Multimodal Integration			
MASM-GNN [77]	✗	✓	✓
SeqGraph-GNN [78]	✗	✓	✓
GPDRP [79]	✓	✓	✓
GraphBAR [80]	✓	✓	✗
DeepMat [81]	✗	✓	✗
BGNN [82]	✓	✓	✗
BGNN (old) [83]	✓	✓	✗

TABLE I: Comparison of different GNN strategies for molecular property prediction, grouped by optimization category.

B. AI for Personalized Drug Discovery and Development

A potential candidate molecule is selected either through manual processes or High Throughput Screening (HTS). However, the number of available drug molecules is so vast that it exceeds the capacity of current laboratory technologies, leaving some areas unexplored. To expedite the drug development process, artificial intelligence has been applied in this field. The most advanced method is GNNs. In this approach, each atom is represented as a node in a graph, and the topology is encoded in the connectivity between these nodes. Although GNNs are widely used, they still face some challenges [84]. The black box still puzzles the field [85]. At present, the prevailing approach is to optimize a model's prediction performance to enhance its explicability. When a model's performance is both stable and excellent, it typically possesses a multilevel architecture [86] that offers valuable clues to help overcome its inherent lack of interpretability by allowing researchers to examine it more thoroughly.

Optimized molecular GNN models are primarily used to optimize molecular properties by leveraging molecular graphs. Within this category, two main strategies can be distinguished. One strategy combines molecular graphs with fingerprints. For example, Cai et al. [70] integrated molecular graphs and fingerprints. The other strategy involves enhancing molecular graph information with additional modules. For instance, Punnachaiya et al. [71] proposed a multimodal GNN that fuses

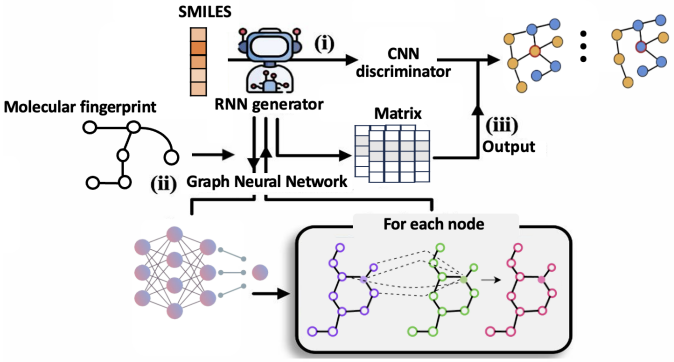


Fig. 3: The figure illustrates two main frameworks for molecular generation: (i) a sequence-based SMILES approach, where RNNs generate molecular sequences and CNNs evaluate them; and (ii) a graph-based approach, where GNNs perform node-level message passing to construct molecular graphs with consistent topology and atomic properties [87]–[91].

molecular graph topology with supplementary modules, such as graph signal processing.

Other optimized molecular GNN models focus on improving the internal model structure. For instance, introducing a new activation function can enhance model accuracy. Rasool et al. proposed Sine Linear Unit [72] to improve GNN prediction results. In addition, key feature and signal information can be combined to further boost performance. Hu et al. proposed a Get Key Clusters (GKC) [73] layer to build a two-stage chemical molecular property prediction model. Another approach involves integrating multiple models for molecular property prediction. Chen et al. [74] fused theoretical computation with the output of a GNN to form a combined model that modulates the final output.

Aside from the black box problem, the field faces additional challenges. One major challenge is generating new molecules with desirable pharmacological, physical, and chemical properties [92]. In other words, even though GNN molecular predictions may appear promising, these molecules often show no activity when synthesized or tested in biological experiments. This is primarily due to inefficiencies in molecular feature extraction. Focusing on molecular properties, the problem can be classified into two categories. One category involves extracting features from molecules—either capturing the entire structure or focusing on local features. Currently, some optimized programs face difficulties extracting features from different molecules due to scale mismatches. One solution is to extract richer biological features. For example, Liu et al. [75] proposed a multi-layer graph network capable of adapting to diverse structural sizes, which improves the robustness and accuracy of predictions.

However, alternative viewpoints also exist. For instance, Wang et al. [76] suggest that most molecular graphs exhibit low clustering along with dominating chains, which lead to topological characteristics that induce feature squashing and thereby reduce the expressiveness of node features. The second category holds that GNN should be integrated with information from other modalities. For example, Zhang et al. [77] use molecular mass spectra as an additional representation, combined with a graph neural network, to provide supplementary information. Similarly, Wang et al. [78] represent molecules as both graphs and sequences, allowing learning from both data types. In another approach, Yang et al. [79] combine two types of data using GNN along with Graph Transformers and deep neural networks. At the same time, GNN technology has entered the protein field. For instance, Xu et al. proposed a GraphBAR [80], a novel deep-learning-based prediction model utilizing graph convolutional neural networks for protein-ligand binding affinity analysis. Currently, more and more new technologies are emerging in this field, including methods based on similarity and feature engineering for predicting compound pathways [81].

Deep learning algorithms are widely used to obtain approximate solutions. For example, Zhang et al. [82] proposed a graph neural network deep learning algorithm as an approximate solution, which can be understood as a representation learning process to extract useful interactions. Similarly, Li et al. proposed a block-based graph neural network (BBGNN) [83], a deep learning algorithm designed to extract useful interactions between a target atom and its neighboring atomic groups.

C. AI in Physiological Signal Analysis for Precision Medicine

Nowadays, Alzheimer's disease remains a major challenge in the field of neuroscience. Even after decades of research, we have achieved significant progress, such as developing new medication therapies. However, the most difficult aspect remains early diagnosis [93]–[96]. The main reason is that the accumulation of amyloid-beta plaques and tau tangles in the brain causes subtle early changes that are difficult to detect [97]. Similarly, Parkinson's disease, a progressive neurodegenerative disorder, also faces several challenges. The loss of a significant number of nigrostriatal dopaminergic neurons and the degradation of striatal terminals are associated with this disorder, and these pathological changes are difficult to identify in the early stages [98]. However, as artificial intelligence advances rapidly, more and more scientists have begun combining AI with medicine. The neural structures and relationships of the brain represent some of the most significant breakthroughs in the field of neurodegenerative disorders.

Electroencephalography (EEG) is one of the most widely used medical technologies in the field of neurodegenerative disorders. EEG still faces many problems. For example, the Short-Time Fourier Transform (STFT) [99]–[101] requires a fixed window width and assumes signal stationarity. Meanwhile, wavelet-based approaches face challenges in selecting appropriate decomposition levels and wavelets, leading to hard-band restrictions in filter banks. However, techniques like signal decomposition and machine learning models are being explored for EEG signal analysis. Deep learning models such as CNNs are applied to classify and detect epilepsy seizures with high accuracy.

1) Electrocardiogram: Deep learning is widely utilized in electrocardiogram (ECG) for early screening of abnormal

cardiac activity [102]. For example, CNNs are highly efficient in forecasting and diagnosing atrial fibrillation (AF). CNNs excel in analyzing complex ECG data, providing higher accuracy than traditional medical machines [103]. However, these models also face difficulty distinguishing certain cardiac diseases, such as ventricular tachycardia (VT) and supraventricular tachycardia (SVT) [104].

Additionally, the sparsity of ECG data has led some researchers to experiment with using only part of the ECG signal as input for deep learning, rather than the entire ECG image. While this approach may reduce computational demands, it can lead to a loss of background information, which may negatively impact the performance of deep learning models [105]. In addition to CNNs, other deep learning technologies are being applied to ECG analysis. For instance, transformer networks have been used for both denoising and segmentation tasks in ECG analysis [106]. As a result, the combination of CNNs and transformer networks has been gaining popularity. For example, Siontis et al. [107] developed a new technology that uses CNNs to extract features from ECG data, while transformer networks convert these extracted features into codes and text strings, improving the output to closely match human performance.

2) Electromyogram: The electromyogram is a bioelectric muscle graph, which plays a significant role in evaluating human activity. For example, sleep bruxism is typically diagnosed using electromyography (EMG) to collect data on jaw muscle contractions and teeth grinding muscle activity [108]. Artificial intelligence is used to analyze this data. Specifically, most research employs CNNs to extract muscle features and investigate the spatial correlations embedded within image representations [109]. Alternatively, CNNs can be used as a decision-making model with attention mechanisms for time-domain and time-frequency domain analysis, followed by feature selection to eliminate redundant features in ECG and other technological data collections [110].

However, to improve the identification ratio, artificial intelligence generally extracts data from raw signals in the time, frequency, and time-frequency domains. The complexity of the data features negatively impacts the system [111]. This issue is compounded by the fact that regular sEMG [112] electrodes have stiff structures, making it difficult to obtain high-fidelity sEMG recordings. One possible reason for this is that, over time, the adhesive properties of the sensors are reduced when they come into contact with the skin surface [113]. Some new research focuses on utilizing AI to automatically learn EMG patterns of attempted movements in paralyzed digits [114] or using electromyography to study facial applications [115].

3) Electroencephalogram: In the field of electroencephalography (EEG), machine learning is the most commonly used artificial intelligence method, especially when compared to other physiological signals. The main reason for this is that machine learning algorithms classify EEG data into segments by focusing on individual EEG signals, thus avoiding the limitations of conventional statistical approaches and often achieving higher accuracy in analysis. Common algorithms for this include transfer learning and adaptive classifiers. Although deep learning is also employed in EEG research, its adoption remains relatively limited [116]. For example, Kurczak et al. [117] created a classifier-based method to analyze EEG data, demonstrating a practical application of machine learning. Other researchers have explored different approaches as well. Sangeetha et al. [118] used a transfer adversarial convolutional architecture to process EEG data, employing reinforcement federated neural networks for feature selection and classification.

Similarly, Wei et al. [119] presented a forest-based approach for analyzing EEG data from sleep spindles. Because artificial intelligence technology is largely data-driven, the multiple comparison problem and the associated statistical corrections can affect both the sensitivity and specificity of analyses [120]. Meanwhile, EEG signals are widely used in neuroscience, and an increasing number of studies are adopting DNN models. However, these models often have limited robustness to noise, leading to reduced precision and increased computational complexity [121]. Furthermore, EEG signal analysis faces additional challenges, including artifacts, long-term recordings, and multiple channels, all of which complicate data processing [122].

As for future directions, because EEG can measure brain activity in real time, it is poised to become the foundation for brain–computer interfaces (BCIs) [123], [124], making this a significant area of emerging research and development.

D. AI for Predicting Treatment Response

Machine learning is a branch of artificial intelligence that has the potential for outstanding performance in disease diagnosis, prognosis prediction, and treatment response evaluation [125]. Machine learning is frequently utilized for predicting treatment responses, as it demonstrates strong performance with relatively small training datasets and is particularly effective at analyzing structured data. Clinical variables—such as patient age, sex, and genetic markers—are primarily structured, thereby enhancing the applicability of machine learning methods in this context. By contrast, deep learning approaches are generally more suitable for unstructured data types, including medical imaging, video, and clinical narratives. As a result, machine learning is often considered the preferred approach for predictive modeling in treatment response studies.

For example, combining machine learning with radiomics has been proven to provide significant medical value in addressing various hepatic diseases [126]. Therefore, it is widely used for medical predictions [127]. Table II provides a detailed overview of machine learning approaches for predicting treatment responses during the 2024–2025 period. In the following, we summarize the key characteristics of machine learning and outline its target tasks.

- Zhang et al. [128] establish a prognostic signature for bladder cancer based on secreted proteins from mesenchymal stem cells, validated via flow cytometry and mFIHC. Using unsupervised clustering and machine learning, the signature effectively distinguishes high- and low-risk patients by correlating MSC presence with tumor characteristics, immune infiltration, and treatment responses.

TABLE II: Overview of machine learning methods for treatment response prediction (2024–2025).

Classification	Method Name	Year	Publication	Algorithm/Model	Feature	Tasks
Machine Learning Cox [128]	MSC-Secretome Signature	2025	Translational Oncology	StepCox plsRcox	MSC-CM signature, TME-related	BLCA prognosis, therapy response
Radiomics ML [129]	Radiomic-MgClassifier	2025	Medical Physics	KNN, RF, SVM, LDA, QDA	OCT texture features, scanner-robust	DME therapy response prediction
Multiomics ML [130]	CMRS	2025	Cancer Cell International	RNA mod types ML ensemble	Multi-cohort, immune	Chemo response, breast cancer prognosis
Multiomics ML [131]	FarmaPRED-PEP	2025	Frontiers in Psychiatry	ML on PGx clinical	Pharmacogenetics, epigenetics, multimodal	Antipsychotic response prediction
EEG [132]	EEG-FC Predictor	2025	Depression and Anxiety	SVM	EEG functional connectivity	SSRI response prediction
fNIRS [133]	fNIRS-ResponseModel	2025	Translational Psychiatry	SVM, RF, XGBoost, LDA, Naïve Bayes	HbT change in dlPFC, nested CV	MDD-treatment response prediction
Comparative [134]	Placebo-MLPredictor	2025	CTS-Clinical and Translational Science	GBM, Lasso, LR, SVM, KNN, RF, ANN	Comparative ML models, placebo propensity	Placebo effect prediction in antidepressant trials
Clinical ML [135]	IO-ML Predictor	2025	JCO Clinical Cancer Informatics	Binary ML Cox model	EHR-based clinical data	ICI response prediction
Radioenomics ML [136]	pLGG-Radiogenomic Model	2025	Nature Communications	Radiogenomic signature	MRI RNA-seq integration	Immune profile and therapy response prediction
Neuroimaging ML [137]	ENIGMA-MDD Predictor	2025	Human Brain Mapping	Gradient, SVM, NN	Cortical morphometry, subcortical volume	Antidepressant response prediction
Radiomics ML [138]	Breast-NAST Predictor	2024	European Radiology	SVM	Ultrasound, tomosynthesis, clinical data	pCR prediction after NAST
Clinical ML [139]	EpilepsyRx-MLPredictor	2024	Epilepsy & Behavior	SVM, KNN, LR, DT, DNN	Clinical chi-square selected variables	Anti-seizure drug response prediction
Radiomics ML [140]	MPM-SVM Predictor	2024	Frontiers in Oncology	CNN feature extractor SVM	CT-based radiomics, double selection	Early response prediction
Network-based ML [141]	PRoBeNet	2024	Journal of Molecular Diagnostics	PRoBeNet biomarkers ML models	Network-informed gene signatures	Therapy response prediction
Multiomics ML [142]	CTR Model	2024	Genes	7 ML algorithms ensemble	CD T cell-related genes	Prognosis and therapy response prediction
Clinical DL (TabNet) [143]	MigraineRx-TabNet	2024	Headache	TabNet SHAP analysis	Clinical questionnaire	Treatment response prediction
Clinical Imaging [144]	CATT-VA Predictor	2024	Ophthalmology Retina	Perceptron NN, Lasso, others	Clinical imaging	VA response prediction to anti-VEGF
Clinical ML [145]	BP-XGBoost Predictor	2024	Nutrition Metabolism	LASSO XGBoost	Clinical features, medications, visit interval	Blood pressure

- Kar et al. [129] develop a radiomic classifier that uses stable, texture-based features from OCT scans to predict anti-VEGF therapy response in diabetic macular edema. By integrating discriminative features from both fluid and retinal tissue compartments, the model achieved an AUC of 0.9, with retinal features demonstrating greater stability.
- Wang et al. [130] present a machine learning-based RNA modification signature derived from 15 breast cancer cohorts. It integrates nine modification types to pinpoint five key prognostic genes and predict survival and treatment response. High CMRS correlates with increased tumor mutation burden and poor cell communication, while low scores suggest improved immune infiltration.
- Mas et al. [131] aim to develop and validate machine learning algorithms that integrate pharmacogenetic, epigenetic, clinical, and neuroanatomical data to classify first-episode psychosis patients based on their antipsychotic treatment response. Leveraging data from the PEPS (N=335) and PAFIP (N=350) cohorts and validating in a new FarmaPRED cohort (N=300), the project addresses key limitations in pharmacogenetic research—such as sample size, phenotype heterogeneity, complexity, and gender perspective—to advance precision medicine in psychiatry.
- Wang et al. [132] used EEG functional connectivity measures from five different conditions to predict treatment efficacy in depression patients treated with escitalopram and sertraline. Employing an SVM classifier, select features demonstrated over 95 percent validation accuracy

and significant correlation with HAMD-17 score reduction, highlighting baseline EEG features as promising predictors of treatment response.

- Ho et al. [133] employ fNIRS and clinical assessments to predict treatment response in MDD using machine-learning models. In a 6-month study of 70 patients, task-induced changes in total haemoglobin in the dlPFC were significantly linked to treatment outcomes, and a Naïve Bayes model using solely fNIRS data achieved a balanced accuracy of 73 percent with an AUC of 0.77, outperforming the combined model.
- Gomeni et al. [134] compared six machine learning methods—including gradient boosting, lasso, logistic regression, SVM, k-nearest neighbors, and random forests—with a multilayer perceptron ANN to predict the probability of individual placebo responses. The ANN achieved the highest accuracy, validated through fivefold cross-validation, and excluding subjects with non-specific effects significantly improved signal detection and effect size.
- Ahluwalia et al. [135] developed a machine learning algorithm using a nationwide electronic health record database of 38,048 advanced NSCLC patients to predict 12-month progression-free and overall survival, outperforming PD-L1 alone. The algorithm achieved AUCs of 0.701 and 0.718 for PFS and OS, respectively, and successfully stratified patients into low- and high-risk groups, with the low-risk group exhibiting significantly lower hazards for disease progression and mortality.
- Fathi Kazeroo et al. [136] integrate MRI and RNA sequencing data from pediatric low-grade gliomas to identify three distinct immune clusters, including one marked by increased immune activity and poorer prognosis that may benefit from immunotherapy. A radiomic signature was developed to predict these immune profiles with over 80 percent accuracy, while a clinicoradiomic model successfully correlated with progression-free survival and treatment response. Genetic variants and transcriptomic pathways associated with tumor growth and immune response were also identified, offering a framework to pinpoint high-risk patients for targeted therapies.
- Poirot et al. [137] applied machine learning to cortical morphometry from 262 MDD patients, but the model did not predict treatment response better than chance (balanced accuracy 50.5 percent), except in extreme responder subpopulations (63.9 percent accuracy).
- Cai et al. [138] developed a model using multi-modal pretreatment ultrasound and tomosynthesis radiomics features combined with clinical variables to predict pCR prior to NAST. In 720 patients, the addition of radiomics features improved performance (AUC from 0.72 to 0.81; FNR 6.7 percent), with surface/volume ratio at tomosynthesis, peritumoral entropy at ultrasound, hormonal receptors, and HER-2 status as key predictors.
- Yildiz et al. [139] conducted a study of 229 patients (aged 1–24 months) in which 11 machine learning techniques, including Support Vector Machine, were evaluated. SVM achieved the highest area under the curve (0.9934) and a test accuracy of 97.06 percent in detecting drug-resistant epilepsy.
- Fanizzi et al. [140] proposed a transfer learning approach to predict initial treatment response from baseline CT scans using three slices as input to a pre-trained convolutional neural network as a radiomic feature extractor. A binary SVM classifier was trained to discriminate responders from non-responders, achieving a median AUC of 86.67 percent and an accuracy of 87.50 percent.
- Shanthamallu et al. [141] proposed a novel framework for precision medicine in autoimmune diseases that prioritizes biomarkers using therapy-targeted proteins, disease-specific molecular signatures, and the human interactome. It reduces features to build robust machine learning models, outperforming models using all or random genes when data are limited.
- Wu et al. [142] developed a CD8 T cell-related prognostic signature consisting of two CTRGs, which are negatively linked to CD8 T cell abundance in the TCGA and METABRIC datasets. The low CTR score group exhibits higher immune infiltration and better responses to chemotherapy and immune checkpoint therapy, with CD163L1 and KLRB1 mRNA downregulated in tumor tissues.
- Chiang et al. [143] applied multiple ML models using clinical and imaging data from the loading dose phase to predict 2-year VA responses in nAMD, achieving an AUC up to 0.87 and R^2 up to 0.45.
- Chandra et al. [144] developed ML models using baseline and weeks 4–12 clinical and imaging data from 1029 CATT participants to predict 2-year VA outcomes in nAMD, achieving AUCs up to 0.87 and R^2 up to 0.45. These results underscore ML's potential to enhance decision-making in anti-VEGF therapy.
- Yi et al. [145] developed a model using 12 phenotypical features from 19,013 visit records (6282 patients) in China to predict individual BP responses. The Extreme Gradient Boosting model achieved a mean absolute error (MAE) of 8.57 mmHg and an R^2 of 0.28, demonstrating significant potential for personalized antihypertensive treatment.
- Liu et al. [146] proposed a federated learning (FL) model using pre-radiotherapy planning target volume images of NSCLC patients to predict CRT treatment response while preserving data privacy. The FL model trains locally and aggregates model parameters, achieving AUCs of 0.725/0.689 (simulation) and 0.698/0.672 (real-world), comparable to centralized deep learning.

Overall, these developments highlight the tremendous potential of machine learning in healthcare. As research continues to expand through multi-omics data and advanced algorithms, machine learning's role in precision medicine is becoming increasingly significant. By leveraging complex data sets to uncover subtle patterns and biomarkers, these approaches enable more accurate diagnoses, tailored treatment strategies, and ultimately, better patient outcomes. This evolution toward data-driven, personalized care is paving the way for a new era in medicine, where innovative computa-

tional tools continuously refine and enhance clinical decision-making.

III. Deep Learning: GNNs, RNNs, and Transformers in medical applications

We present representative hierarchical neural network architectures widely adopted in precision medicine, along with their historical development and key advantages. Taking the neurocognitive machine proposed by Fukushima et al. [147] as an example, its main contribution lies in defining a multi-layer neural network architecture composed of the G, S, and C layers. Specifically, the G layer processes the input image, the S layer extracts its features, and the C layer corrects any errors in the representation. This structure offers an intuitive and early illustration of how neural networks function.

This architecture later served as the foundation for CNNs. A typical CNN consists of four main components: a convolutional layer that scans the image, a pooling layer that removes redundant details, a fully connected layer that infers the content based on extracted features, and an output layer that generates the final prediction. CNNs differ from traditional fully connected neural networks. They apply convolutional filters that slide over the image to detect local patterns. This sliding mechanism enables CNNs to capture spatial relationships between neighboring pixels, thereby improving their understanding of image structures. In contrast, fully connected neural networks process raw pixel values independently and lack the ability to model spatial dependencies [148]. In CNN, each pixel is considered as a feature; thus, this results in image datasets having high dimensionality. Generally, convolutional neural networks can reduce dimensionality without sacrificing the images' essential characteristics. The robustness of CNN makes it suitable for image and visual processing problems [149]. The main advantage of CNNs lies in extracting part of the features and data information. However, CNNs do not always have perfect performance. Recurrent Neural Networks (RNNs) are similar to CNNs, as they are both part of deep learning. RNN models are suitable for sequential data and tasks where the order of the data points is crucial, such as text processing and time series analysis. The main reason is that RNNs learn through Backpropagation Through Time (BPTT) to transfer error information to the earlier layers, which can help achieve accurate results. Furthermore, RNNs can be easily extended to two areas: Recursive Neural Networks and Graph Neural Networks [150]. But they struggle with long sequences due to the vanishing gradient problem [151]. Meanwhile, both models face similar challenges. All artificial neural networks, including RNNs and CNNs, can only identify objects that belong to the classifications they have been trained on. Therefore, if the user inputs data that does not belong to the training set, the model will not output an accurate result [152]. However, since these results were mostly produced in scenarios with plenty of training data, it is not apparent whether they can be immediately applied to low-resource circumstances [153].

Transformer was first used for natural language processing and gradually became a key model in this field. Its main function is to move the generalization capability of old patterns to new ones. Inductive learning and transductive learning are among the components of transfer learning [154]. But the good performance of Transformer requires pretraining on large datasets, which limits the application of pure Transformers to medical image datasets with smaller amounts of data [155].

A. Graph Neural Networks: Biological Pathways

Biological pathways are a series of reactions in biological systems, including various kinds such as metabolism, cellular processes, and human diseases. These pathways are crucial for cellular processes such as disease progression, metabolism, the immune system, and signal transduction, and are widely used as prior knowledge in medical research [156], [157]. For example, cancer cells utilize the internal environment to release growth factors and cytokines, which activate surrounding normal cells, initiating rapid cancer cell duplication [158]. Therefore, identifying specific genes (biomarkers) is crucial because they serve as indicators of how drug responses are influenced [159].

In the real world, linear fitting explains many phenomena; however, in biological systems, the internal complexity of gene interactions is almost always better described by nonlinear fitting, making conventional statistical analysis invalid [160]. At the same time, disease models can be identified and classified into general methods and specific disease methods. The general method aims to uncover disease models based on the molecular topology network structure. The powerful functionality of GNNs can optimize the general method, as GNNs have been used to capture network nodes and combine node properties to achieve superior results [161].

GNNs are a type of deep learning method that has been increasingly applied in recent bioinformatics research. For example, Ren et al. [162] mention that GNNs have gained wide attention for their ability to capture topological information, with a key function of discovering predictive factors and key genes that reflect the immunotherapy response and prognosis of bladder cancer. GNNs offer a promising avenue for Autism Spectrum Disorder (ASD) by integrating multi-dimensional biological data and exhibiting immense potential [163]. Currently, the field of computer science faces many research gaps. For example, model construction often requires biological knowledge to design experiments [164]. Drug interactions remain unclear when patients take medication [165], as the components of the drug and their interactions with proteins can change. Additionally, most models fail to consider the topological structure of pathways and instead treat pathways merely as a set of genes. Moreover, many interactions between genes and proteins remain undiscovered, and immune cell infiltration levels are often overlooked [166], [167].

B. Reinforcement Learning: Optimizing Personalized Treatment Regimens

Reinforcement Learning (RL), as a type of adaptive mechanism under machine learning, enables software agents to learn and update themselves dynamically through trial-and-error by taking actions in their environment. Unlike other computational approaches, RL allows agents to learn from

interacting with their environment without explicit supervision or a complete environmental model. Typically, the agent's actions result in both immediate and delayed rewards, leading the agent to identify the optimal next action within the state space. The agent operates in a trial-and-error manner to maximize cumulative rewards over time. In the standard RL framework, the role of RL in chemotherapy can be described using a Partially Observable Markov Decision Process (POMDP) [168], [169]. Broadly, RL methods involve learning through trial and error to train agents that select actions maximizing a reward function. Importantly, RL methods are particularly suited for solving optimization problems where the dynamics of the system are unknown, as they do not require predefined system models [170], [171].

Reinforcement learning–based algorithms do not efficiently learn from physicians when applied to drug treatment decisions. Moreover, these models do not fully represent the patient state space and action space, which limits their performance [172]. On the other hand, reinforcement learning lacks stability and does not guarantee optimal medical treatment. For example, sepsis clustering methods depend on the initial parameters and sometimes yield different results [173].

For example, Ghasemi et al. [174] applied off-policy reinforcement learning and used weighted importance sampling for evaluation. At the same time, research by Drudi et al. [175] employed a policy iteration algorithm to obtain the optimal AI policy. Additionally, Abebe et al. [176] proposed decision tree-based reinforcement learning approaches. Although these three approaches differ, they all fundamentally aim to optimize strategic decision-making. Other methods consider using hybrid models to optimize strategies. For instance, Zhang et al. proposed DDQN [177] with expert strategies to optimize medication decisions.

The second approach is reward formulation optimization. Reinforcement learning's key components are agent, environment, state, action, reward, and policy. Reward is the most critical aspect because it guides the agent through effective trial-and-error learning, enabling sequential decision-making and enhancing performance [182].

Regarding reward optimization, current research priorities fall into distinct categories. One category focuses on the rewards process, such as Guo et al., who proposed a novel RL-HMM (hidden Markov model) [178] framework for analyzing reward-based decision-making. Another category focuses on defining and formalizing the reward function. For example, Shah et al. [169] developed a method to learn the reward function directly from data obtained through experimental trials, thereby addressing the challenge of reward function definition. Similarly, in addition to acquiring data from experimental trials, clinical data can also be used for training. For instance, Lu et al. [179] developed an interpretable reward formulation to guide the Deep Reinforcement Learning (DRL) agent in learning from real clinical data. Reinforcement learning for automatic planning is considered one of the most promising applications, as it can mimic the decision processes of an expert planner [183].

Recently, new reinforcement learning hybrid models have emerged. For example, Nambiar et al. refined traditional off-policy RL methods to address action imbalance during offline training [180]. Zhang et al. [181] proposed a constant force-tracking control strategy for dynamic environments, along with a modeling approach that adheres to physical constraints.

C. Generative AI: Diffusion Models and GANs for Molecule Synthesis

Generative artificial intelligence is a branch of AI primarily focused on generating text, images, videos, voices, and more. In medicine, it is used for molecule generation, with common technologies including diffusion models and Generative Adversarial Networks (GANs) [184], [185].

1) Diffusion Models: Diffusion models are a new type of generative model that have been widely applied in image super-resolution [186], [187], hair transfer [188], sign pose generation [189], hand pose estimation [190], human interaction generation [191], robot navigation [192], and molecular design [193]. Taking the Di BindFR [194] diffusion model as an example, most workflows that use diffusion models for molecular docking are generally similar to that of the Di BindFR model. Diffusion models (DMs) are key to advancing computational structural biology. This technology originated in computer vision and is rapidly transforming image generation. It was first introduced in 2015 by Alakhdar et al. [195] and achieved a breakthrough in image generation in 2020.

DMs are well-suited for molecular synthesis for several reasons: (i) They do not require precise conditional training, which provides flexibility; (ii) This technology has been successfully applied in the protein field, demonstrating its capability to model complex protein data; and (iii) Their multi-step process allows for the gradual refinement of desired properties, and they can be effectively combined with variational autoencoders (VAEs) [196]. Diffusion models operate using two main strategies. The first is the exploration phase, during which the model selects the atomic species. The second is the relaxation phase, where the model identifies a low-energy geometry [197]. At present, there are three main types of models. The first type focuses on protein drug exploration. For example, Xu et al. [198] integrated forward and backward diffusion processes to control the generation gradient of conformational states, targeting P-type ATPases—a critical family of membrane transporters. The second type focuses on denoising, as the denoising diffusion probabilistic model is a highly efficient approach for addressing the lack of diversity. For example, Cao et al. [199] proposed a three-stage text-guided conditional denoising diffusion probabilistic model to enhance diversity in antimicrobial peptides.

The third type focuses on graph diffusion models [200], [201]. These models corrupt discrete data using a transition matrix to map it into a continuous space for the diffusion process, which can make extensible conditional generation challenging. Bian et al. introduced the Hierarchical Graph Latent Diffusion Model (HGLDM) [202] to address this issue. Other types of models address molecular challenges. For example, Wang et al. [203] proposed a transferable graph attention diffusion model to enhance molecular conformation noise prediction. Additionally, Wang et al. introduced the

TABLE III: Summary of recent reinforcement learning strategies applied to healthcare decision-making.

Method Name & Citation	Optimization Focus	Strategy / Description
Weighted Off-policy RL [174]	Strategy Learning	Importance sampling to improve off-policy evaluation
Policy Iteration RL [175]	Strategy Learning	Iterative learning of optimal decision policies
Decision Tree RL [176]	Strategy Learning	Rule-based decisions via decision trees
Safe-DDQN with Expert Injection [177]	Strategy Learning	Combines dueling DDQN with expert knowledge
RL-HMM [178]	Reward Modeling	Hidden Markov model to analyze reward mechanisms
Data-driven Reward Learning [169]	Reward Modeling	Reward function inferred from experimental trial data
Interpretable Reward DRL [179]	Reward Modeling	Clinical data-based reward formulation
Offline Balanced Deep RL [180]	Hybrid Architecture	Addressing action imbalance in offline learning
Force-Tracking Control RL [181]	Hybrid Architecture	Combines dynamic control strategy with physical constraints

Graph Latent Diffusion Model (GLDM) [204], which uses autoencoders to compress complex chemical data and leverages diffusion model capabilities to generate novel molecules. Moreover, Guo et al. [205] presented a novel approach for generating molecules with desirable properties by employing a mixed Gaussian distribution in the reverse diffusion process.

In summary, various diffusion models have been developed to tackle molecular challenges through different approaches. By leveraging techniques such as graph attention, autoencoders, and mixed Gaussian distributions, these approaches demonstrate the versatility of diffusion models in processing complex chemical data. Due to the complexity of chemical space, researchers currently focus on generating small molecules [206]. Moreover, diffusion models exhibit a generative process that is random and fails to incorporate atomic interaction information between ligands and proteins [207]. Additionally, existing diffusion models typically suffer from high modeling complexity and inadequate coverage of multimodal distributions [208]. In the field of generative artificial intelligence, a new development has emerged: the use of diffusion models for 3D molecular modeling. This innovative approach leverages diffusion processes to generate detailed and accurate three-dimensional molecular structures, paving the way for advanced drug discovery and design [209], [210].

2) Generative Adversarial Networks (GANs): Generative Adversarial Networks (GANs) are deep learning models that can learn from data and generate new samples [211]–[217]. However, GAN models ignore the fact that each molecule exhibits specific stability states across multiple stable conformations. As a result, generative artificial intelligence often struggles to capture long-range dependencies between atoms [218], much like how PPI [219] interfaces and compounds targeting PPIs display distinct physicochemical properties compared to traditional binding pockets and small-molecule drugs. Moreover, many molecular models rely solely on processing input information in a forward manner [220], which can lead to incomplete molecular development due to errors [221]. These factors not only harm the structural accuracy of the generated molecules, but also significantly impact overall model accuracy.

Currently, new research has introduced several methods to improve model accuracy. For example, Anoshin et al. [222] integrated parameterized quantum circuits into existing molecular generative adversarial networks to enhance performance and stability during training. Additionally, Zhao et al. proposed and evaluated a new GAN, the Language Model Guided Antibody Generative Adversarial Network (AbGAN-LMG) [223], which uses a language model as input to improve the generation of high-quality molecules. Building on these efforts, another strand of research has focused on enhancing the transferability of these models. For example, Janson et al. [224] developed a transformer-based generative adversarial network that achieves improved transferability beyond the constraints of limited training data.Additionally, Fan et al. [225] demonstrated that a Vanilla GAN architecture can generate valid molecules and further explored how incorporating a reward network within a MolGAN framework influences the validity score under different scenarios.

These studies reveal that enhancing transferability is crucial for improving the generalization and robustness of generative models in molecular design. Beyond molecular applications, GANs also show high potential in broader data generation fields, because when researchers target specific protein targets, they often have limited pharmaceutical data available for training. For example, Aphikulvanich et al. [226] proposed a Bayesian optimization approach to determine the optimal singular values of model weights, enabling the generation of drug molecules with ideal properties even without massive amounts of data.

However, GANs are not a one-size-fits-all solution. Different applications require unique frameworks, tailored model training strategies, and specific hyperparameter configurations to develop a suitable model, often necessitating extensive adjustments.

D. Multi-Modal Learning: Imaging

Multimodal image fusion is a critical task in precision medicine. For instance, in brain imaging, multimodal learning that integrates multiple images plays a vital role in clinical applications [227]. Typically, this process involves merging two or three images into a single composite image, thereby providing healthcare professionals with supplementary information for clinical decision-making.

The image fusion process involves extracting features from images. Deep learning approaches that integrate multimodal datasets typically extract more comprehensive features than those using unimodal datasets [228]. However, effective feature extraction requires capturing either local or global characteristics. Commonly employed methods for this task include

Multidimensional Scaling (MDS) [229] and Sparse Representations [230].

1) Multidimensional Scaling (MDS): Multidimensional Scaling (MDS) operates by rescaling image dimensions while preserving the intrinsic spatial relationships among data points, thereby enabling computational models to capture image features more effectively. Among the diverse techniques employed in MDS, one notable approach is the Laplacian Pyramid. In practice, this technique first applies a Gaussian blur to reduce fine details, producing a low-pass filtered image for feature extraction. By iteratively repeating this process, a pyramid-like structure is formed that facilitates the extraction of increasingly abstract features across multiple scales [231]. However, MDS lacks inherent intelligence, as its operation is primarily based on pixel-level abstraction and does not automatically discern which features are important. With advancements in ANNs, novel approaches are emerging that integrate MDS with these networks. Relying solely on ANN-based methods has led to challenges, such as the issues encountered by CNNs with imbalanced classes, blurred decision boundaries, and high parameter requirements [232]. By combining these two technologies, MDS can be used to extract multidimensional representations, while CNNs perform automated feature learning to identify critical features, ultimately enhancing image resolution and yielding high-quality images [233], [234].

2) Transformer Models: Vaswani et al. [243] proposed the Transformer model, which has rapidly transformed the field of deep learning—not only by its inherent design but also through its applicability in multimodal data fusion. The most significant breakthrough of this model is the use of a self-attention mechanism as its core architecture, eliminating the need for RNNs and enabling more effective integration of diverse data modalities. Since its introduction, the Transformer model has become the dominant architecture in deep learning, serving as the foundation for numerous subsequent studies and model innovations, particularly in the realm of multimodal applications [244]–[251].

Leveraging a self-attention mechanism, the Transformer model is frequently trained using self-supervised or unsupervised learning paradigms, which help avoid the labor-intensive process of manual data annotation [252]. This approach is especially beneficial for fusing heterogeneous data sources, as it allows the model to automatically learn complementary features across various modalities. The Transformer model has proven effective in processing and interpreting diverse data, such as images and structured data; consequently, numerous transformer-based models have been proposed for healthcare applications where multimodal data fusion is critical [253]. Moreover, owing to its advanced token processing capabilities, the transformer is extensively employed for analyzing long sequential data, including medical images and electronic medical records (EMRs) [254].

There are various types of data fusion, and one type involves integrating data from long sequence processing. For example, Nguyen et al. [236] proposed two transformer-based components. One component is designed to analyze image data, while the other utilizes the output from the image analysis, fusing it with clinical data to predict changes in knee osteoarthritis and the clinical status of Alzheimer's disease. This example underscores how the Transformer's architecture is particularly well-suited for combining heterogeneous data types to improve diagnostic accuracy and patient outcomes. Recently, considerable progress has been achieved in overcoming the challenges associated with multimodal data fusion. This table summarizes current representative solutions for multimodal fusion across different modeling strategies IV. The first approach is based on Transformer/attention mechanisms for multimodal fusion. For example, Zhao et al. proposed the cross-modal feature fusion (CMFF) method [235] for modal feature fusion, which uses multi-head attention mechanisms to obtain informative feature representations and improve diagnostic accuracy.

Nguyen et al. [236] proposed a method for predicting prognosis using two Transformer-based components that share information. The first Transformer analyzes the imaging data, while the second fuses the clinical data. Wang et al. proposed a Multi-Feature Optimization Transformer (MFOT) [237] to encode visual grid features from different dimensions, thereby enhancing interaction capabilities. Collectively, these studies demonstrate that a flexible framework can dynamically fuse heterogeneous features—ultimately driving multimodal fusion. The second approach is based on graph structures for multimodal fusion.

For example, Zhang et al. [238] collected molecular graph data and spectra from a multimodal dataset for data augmentation. Ma et al. [239] proposed addressing this issue by using a flexible and interpretable graph structure to fuse and model multimodal data, thereby overcoming fragmentation and one-sidedness. Other studies have utilized knowledge graph techniques for data fusion. Gao et al. [240] employed multi-modal knowledge graph node types, text, and numerical information for feature fusion. The third approach is based on single-modal feature extraction for multimodal fusion. Kanyal et al. [241] proposed a deep learning model that extracts features from structural magnetic resonance imaging (sMRI), functional magnetic resonance imaging (fMRI), and genetic markers such as single nucleotide polymorphism (SNP), and then fuses these features together. Yang designed a multimodal object-level contrast (MOC) [242] learning method that operates in conjunction with heterogeneity reduction to optimize multimodal cancer survival prediction datasets.Collectively, these approaches illustrate the current landscape of multimodal fusion methods. While these approaches offer robust frameworks for multimodal fusion, several aspects remain untapped and warrant further exploration.For example, only a few studies have explored multimodal fusion involving X-rays, echocardiography, non-imaging modalities, or the cardiovascular domain [255].

In summary, it is important to note that complex multimodal data fusion creates a search space prone to local minima, which hinders the attainment of optimal solutions. This remains a challenge that needs to be addressed in future research [256]. Alternatively, extensive multimodal data can be collected to investigate the dynamic interplay of symptoms, severity, triggers, and modifying factors [257].While the feature extraction technology in multimodal image fusion has

TABLE IV: Comparison of multimodal fusion strategies categorized by modeling approach.

Category	Method / Reference	Solution Summary
Transformer-based Fusion		
CMFF [235]	Cross-modal feature fusion using multi-head attention	Improves diagnostic accuracy through enhanced feature representation
CLIMATv2 [236]	Dual transformer components	Shares modality-specific information for prognosis prediction
MFOT [237]	Visual grid encoding from multiple dimensions	Enhances feature interaction and fusion capacity
Graph-based Fusion		
TransG [238]	Multimodal graph and spectra data fusion	Facilitates data augmentation using structured graph representation
FGCN [239]	Interpretable graph structure	Fuses fragmented and one-sided multimodal data
CGAT-ADM [240]	Multimodal knowledge graph with text and numeric nodes	Integrates heterogeneous features for robust fusion
Feature Extraction for Fusion		
XGBoost Fusion Pipeline [241]	Feature extraction from sMRI, fMRI, and SNP	Deep learning model that fuses features for prediction
MOC [242]	Multimodal object-level contrastive learning	Reduces modality heterogeneity in survival prediction

matured, current methods do not fully capture multidimensional features [258], particularly as data volumes continue to increase. Consequently, multimodal image fusion models now face the challenge of effectively capturing contextual information [259]. Achieving a breakthrough in enabling machines to learn and utilize contextual cues may well usher in the next technological revolution, leading to significantly improved model performance and more robust applications.

IV. DISCUSSION

AI has numerous applications. This review provides a brief history of research achievements in major areas, highlighting examples of research and development as well as the evolution of techniques. In the field of AI, the USA, Canada, the UK, and China are experiencing rapid development [260]–[262]. Currently, artificial intelligence technologies have been widely applied across diverse domains: augmented reality (AR) enhances visualization performance; biometric identification improves user authentication and access control through behavioral data [263]; and in biomedicine, advanced imaging combined with machine-learning and deep-learning methods enables accurate risk stratification to support clinical diagnosis [264].

As we navigate this era of rapid technological advancement, it is imperative to ground AI deployment in sound principles [265]. AI should serve as a supplement in real-life applications, with an essential focus on providing auxiliary services rather than replacing human expertise—there is no substitute for direct human involvement [260], [266]. However, supporting both routine operations and critical-care scenarios requires robust infrastructure—including reliable data storage, high-speed networks, and comprehensive security measures—alongside strict data-privacy standards and international compliance [267]. At the same time, sustainable and inclusive development must be prioritized to ensure these innovations are deployed responsibly [268].

Unwanted data breaches heighten security concerns, making it necessary to address data protection in every context. It is equally crucial to eliminate biases from data and algorithms and to establish clear regulations to govern AI use [269]. Moreover, by involving ethics scholars who embody the principles of justice, fairness, and social responsibility, we can continuously improve ethical and legal frameworks, thereby expanding AI developers' knowledge and fostering responsible innovation [270]. Recent surveys indicate that although many people have limited experience with artificial intelligence, there is strong interest in learning about AI. This suggests an urgent need to develop education and training programs that equip diverse professionals with AI skills [271]. Future research and practice should focus on leveraging AI to boost productivity, enhance task quality, and reduce completion times—for example, by automating coding tasks or generating reports [272].

However, widespread AI adoption also introduces significant risks. Overreliance on pattern recognition can amplify existing biases in data, leading to discrimination or fraud [273]. In healthcare, clinicians express concern that unchecked AI use could contribute to medical negligence. To mitigate these dangers, clear ethical guidelines and robust regulatory frameworks are essential. Governments must therefore establish comprehensive policies, standards, and oversight mechanisms to ensure that AI technologies are developed and deployed responsibly, equitably, and safely [274], [275].

V. CONCLUSION

In this work, we present an overview of the current landscape of precision medicine, highlighting key AI-driven advances and persistent challenges. We summarize core technologies such as deep learning and machine learning, along with their clinical applications. We hope to inspire future research, cross-disciplinary collaboration, and practical innovation in precision medicine. By fostering collaboration across disciplines, we aim to accelerate the translation of AI innovations into clinical practice. Together, these efforts will help create an environment in which AI can meaningfully

enhance patient outcomes and support the future of precision medicine.

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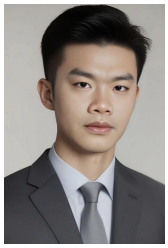
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Zhi Xing Wang is currently pursuing the bachelor's degree at the University of Malaya, Malaysia. He is a research assistant with the team of Prof. Hao Tang at Peking University, China. His research interests lie in the intersection of artificial intelligence and biomedical science, with a particular focus on applying AI techniques to address problems in healthcare and life sciences.



Hao Tang is a tenure-track Assistant Professor at Peking University, China. Previously, he held postdoctoral positions at both CMU, USA, and ETH Zürich, Switzerland. He earned a master's degree from Peking University, China, and a Ph.D. from the University of Trento, Italy. He was a visiting Ph.D. student at the University of Oxford, UK, and an intern at IIAI, UAE. His research interests include AIGC, LLM, machine learning, computer vision, embodied AI, and their applications to scientific domains.